

WEEK 13: File & Text Processing (shell scripting)

1. script to count the number of lines, words, and characters:

```
MINGW64 ~/C/Users/raako/raa x + v
```

```
#!/bin/bash

file="devops_count.txt"
lines=$(wc -l < $file)
words=$(wc -w < $file)
characters=$(wc -c < $file);

echo "File name : $file"
echo "Lines: $lines"
echo "Words: $words"
echo "Characters: $characters"
```

```
count.sh [unix] (01:08 09/02/2026) 3,22 All
"count.sh" [unix] 11L, 211B
```

25°C Mostly clear Search 01:11 09-02-2026

Output:

```
raake@LAPTOP-049ULBK0 MINGW64 ~/raakdevops/week13_project
$ ./count.sh
file name : devops_count.txt
Lines: 3
Words: 25
Characters: 154:
```

- ## 2. script to replace all occurrences of the word "DevOps" with "CloudOps"

```
raake@LAPTOP-049ULBK0 MINGW64 ~/raakdevops/week13_project
$ cat replace.sh
#!/bin/bash

file="devops_count.txt"

sed -i 's/Devops/cloudops/g' "$file"

echo "Devops is changed as cloudops"
```

Output:

```
raake@LAPTOP-049ULBK0 MINGW64 ~/raakdevops/week13_project
$ ./replace.sh
Devops is changed as cloudops
```

CRON JOBS:

Scheduling a cron job to run your backup script daily at 2 AM.

```
LAPTOP-049ULBK0:~/raakdevops# crontab -l
# do daily/weekly/monthly maintenance
# min    hour    day    month    weekday  command
*/15     *      *      *      *        run-parts /etc/periodic/15min
0        *      *      *      *        run-parts /etc/periodic/hourly
0        2      *      *      *        run-parts /etc/periodic/daily
0        3      *      *      6        run-parts /etc/periodic/weekly
0        5      1      *      *        run-parts /etc/periodic/monthly
02***   /c/users/raakdevops/cron_test.sh
```

Exported to mycronjob.txt

```
LAPTOP-049ULBK0:~/raakdevops# crontab -l > mycronjob.txt
LAPTOP-049ULBK0:~/raakdevops# cat mycronjob.txt
# do daily/weekly/monthly maintenance
# min    hour    day    month    weekday  command
*/15     *      *      *      *        run-parts /etc/periodic/15min
0        *      *      *      *        run-parts /etc/periodic/hourly
0        2      *      *      *        run-parts /etc/periodic/daily
0        3      *      *      6        run-parts /etc/periodic/weekly
0        5      1      *      *        run-parts /etc/periodic/monthly
02***   /c/users/raakdevops/cron_test.sh
```

Learning from shell scripting and automation:

In this, I learned the fundamentals of shell scripting and how automation is used in DevOps to simplify and manage daily operational tasks. I understood that shell scripting allows us to combine multiple Linux commands into a single script so that tasks can be executed automatically without manual effort. This helps reduce human errors and saves a lot of time, especially when tasks are repetitive.

I learned how to create shell script files using the .sh extension and the importance of the shebang line (`#!/bin/bash`), which specifies the shell interpreter. I also learned how to give execution permission to scripts using the `chmod +x` command. This helped me understand how scripts are treated as executable programs in a Linux environment.

Through this project, I gained hands-on experience in file and text processing. I learned how to count the number of lines, words, and characters in a file using the `wc` command. This is useful when working with log files or large text files in real systems. I also learned how to replace specific words in a file using the `sed` command, which is a powerful tool for automated text editing. This showed me how scripts can modify files quickly without opening them manually.

Another important concept I learned was automation using cron jobs. Cron jobs allow scripts to run automatically at scheduled times, such as daily, hourly, or weekly. I learned how to use `crontab -e` to define schedules and how cron helps in running scripts even when no user is logged in. This made me understand how DevOps engineers rely on automation to maintain systems continuously.

I also learned the importance of logging and output files. By redirecting script output to text files, it becomes easy to verify whether a script ran successfully. This is useful for debugging and monitoring automated tasks. Overall, this project helped me understand how scripting and automation form the backbone of DevOps by improving efficiency, consistency, and reliability in system operations.

Real-World Scenario Where Shell Script Is Used in DevOps

One real-world scenario where shell scripting is commonly used in DevOps is **automated log file management on production servers**. In real-time applications, servers generate large log files every day that record application activities, errors, and system events. If these log files are not managed properly, they can consume disk space and lead to system failures or application downtime.

In such cases, a DevOps engineer can write a shell script to automatically identify log files older than a certain number of days, for example 7 or 14 days, and delete or archive them. This script ensures that only recent and relevant logs are kept on the server. The script can also compress older logs to save disk space instead of deleting them completely.

This shell script can be scheduled using a cron job to run daily during non-peak hours, such as late at night. Automation ensures that log cleanup happens regularly without manual intervention. This reduces the risk of disk space exhaustion and improves system stability. It also frees the DevOps team from performing repetitive maintenance tasks manually.

Apart from log management, similar shell scripts are used in real-world DevOps for automated backups, disk usage monitoring, service health checks, and deployment support tasks. For example, a script can check disk usage and alert the team if usage crosses a threshold, or restart a service automatically if it stops unexpectedly.

In real DevOps environments, even when advanced tools like Jenkins, Ansible, or Docker are used, shell scripts still play an important role behind the scenes. This makes shell scripting a fundamental and essential skill for DevOps engineers.